Task 10 Report: AI vs ML vs Deep Learning

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GitHub: https://github.com/sanzayjanarthanan/ai-vs-ml-vs-deep-learning

1. Objective

To build a clear understanding of the relationship between Artificial Intelligence, Machine Learning and Deep Learning using definitions, differences and real-world examples.

2. Definitions

• **AI:** The broader field of building machines that perform tasks that normally need human intelligence, such as reasoning, understanding language and decision making.
• **Machine Learning:** A subset of AI where systems learn patterns from data and improve without being explicitly programmed with rules.
• **Deep Learning:** A subset of ML that uses multi-layer neural networks to learn complex patterns from large amounts of data.

3. Comparison Table

Feature	AI	Machine Learning	Deep Learning
Scope	Broadest field	Subset of AI	Subset of ML
Goal	Mimic human intelligence	Learn patterns from data	Learn complex patterns using neural networks
Data needed	Varies (can be rule-based)	Small to medium	Very large
Feature engineering	Depends on approach	Often manual	Learned automatically
Compute needed	Low to high	Moderate	High (GPU/TPU)
Interpretability	Varies	Often easier to explain	Harder (black box)
Example algorithms	Search, rule-based systems	Linear regression, decision trees, SVM	CNN, RNN, Transformers

4. Real-World Examples

• **AI:** Chess engines, voice assistants, rule-based chatbots
• **ML:** Spam email filter, recommendation systems, house price prediction
• **Deep Learning:** Image recognition, speech recognition, face unlock, large language models, sign language recognition

5. How They Are Related

AI is the big umbrella, ML sits inside AI, and Deep Learning sits inside ML: **AI > Machine Learning > Deep Learning**. Every deep learning model is a machine learning model, and every machine learning model is AI. But not all AI is ML (e.g. a hand-written rule-based system), and not all ML is deep learning (e.g. a decision tree).

6. Conclusion

Choose the approach based on the problem: simple rules for simple tasks, ML for learning patterns from structured data, and Deep Learning when data is large and unstructured (images, audio, text).